Problem Set 2

Applied Stats/Quant Methods 1

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

Answer: The chi-square test statistic is:

$$\chi^2 = 3.791, \quad df = 2$$

Computed in R as:

```r
##Question1a: Calculate Chi-square manually

#storing the counts into a vector

counts <- c(14, 6, 7, 7, 7, 1)

#naming rows and columns
row_labels <- c("Upper class", "Lower class")
col_labels <- c("Not Stopped", "Bribe requested", "Stopped/given warning")

#creating the contingency table
contingency_table <- matrix(data = counts, nrow = 2, ncol = 3, byrow = TRUE, dimnames = list(Class = row_labels, Outcome = col_labels))
table_with_margins <- addmargins(contingency_table)

#extracting the totals
row_totals <- table_with_margins[, "Sum"][1:2]
col_totals <- table_with_margins["Sum", ][1:3]
grand_total <- table_with_margins["Sum", "Sum"]

#looping through calculating expected value for each cell
expected_values <- matrix(NA, nrow = 2, ncol = 3, dimnames = dimnames(contingency_table))

for (i in 1:nrow(expected_values)) {
  for (j in 1:ncol(expected_values)) {
    expected_values[i, j] <- (row_totals[i] * col_totals[j]) / grand_total
  }
}
print(expected_values)

#calculating Chi-square
chi_square <- sum(((contingency_table - expected_values)^2) / expected_values)
```

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

Answer: The p-value obtained was $p = 0.150$, which is greater than $\alpha = 0.1$. We therefore *fail to reject* the null hypothesis and conclude that there is no statistically significant association between social class and outcome of the stop.

Computed in R as:

```
##Question 1b: computing p-value

#calculating p-value
df <- (nrow(contingency_table) -1) * (ncol(contingency_table) - 1)
alpha <- 0.1
p_value <- 1 - pchisq(chi_square, df = df)
p_value
is_significant <- alpha > p_value
```

[2] Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.136	-0.815	0.819
Lower class	-0.183	1.094	-1.099

Computed in R as:

```
##Question 1c: computing standardised residuals
chi_test <- chisq.test(contingency_table)
residuals <- chi_test$residuals
residuals
```

(d) How might the standardized residuals help you interpret the results?

Answer: If the Chi-square test were significant (which it was not in this case), we would know that the variables are related, but not which specific categories drive that relationship. The standardised residuals address this by showing how much each observed count deviates from its expected count, measured in units of standard deviation.

Values close to zero indicate that the observed count is very similar to what would be expected under the null hypothesis of independence, whereas residuals greater than $|2|$ are typically considered large enough to indicate a significant deviation. In our case, none of the standardised residuals exceed $|2|$, which confirms the earlier finding that the Chi-square test was not statistically significant at $\alpha = 0.10$.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure **??** below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

Answer: The null hypothesis assumes that there is *no difference* between villages with reserved seats for women and those without. The alternative hypothesis assumes that the reservation policy *does have an effect*, meaning there is a difference, but it does not specify the direction of that difference.

$$H_0: \ \mu_{\text{reserved}} = \mu_{\text{not reserved}} \quad \text{vs.} \quad H_a: \ \mu_{\text{reserved}} \neq \mu_{\text{not reserved}}$$

(b) Run a bivariate regression to test this hypothesis in `R` (include your code!).

Answer: A bivariate regression was estimated to test whether villages with reserved seats for women differ from those without in the outcome variable.

	Dependent variable:
	Outcome
Reserved (1 = Yes)	9.252**
	(3.948)
Constant	14.738***
	(2.286)
Observations	322
R^2	0.017
Adjusted R^2	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
Note:	*p<0.1; **p<0.05; ***p<0.01

Computed in `R` as:

```r
##Question 2b: Building a model

#load data
data_url <- "https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv"
west_bengal_data <- read.csv(data_url)

#scatterplot
plot(west_bengal_data$reserved~west_bengal_data$water,main='Women in power and policy differences',xlab='reservation policy',ylab='number of new or repaired drinking water facilities',pch=20)

#build model
model <- lm(water ~ reserved, data = west_bengal_data)
summary(model)
```

```

# producing LaTeX code
install.packages("stargazer")
library(stargazer)

stargazer(model, type = "latex",
          title = "Bivariate Regression of Outcome on Reservation Policy",
          dep.var.labels = "Outcome",
          covariate.labels = c("Reserved (1 = Yes)"),
          digits = 3,
          no.space = TRUE)
```

(c) Interpret the coefficient estimate for reservation policy.

Answer: The estimated intercept, $\hat{\beta}_0 = 14.74$, represents the average number of new or repaired water facilities in villages where the seat was *not* reserved for women. The coefficient for the reservation policy, $\hat{\beta}_1 = 9.252$, captures the estimated causal effect of the policy. It indicates that villages with a reserved seat had, on average, about 9.25 more new or repaired water facilities than those without a reserved seat.

The p-value associated with $\hat{\beta}_1$ is $p = 0.0197$, which is below $\alpha = 0.05$. We therefore *reject the null hypothesis* that the reservation policy had no effect. The policy is associated with a statistically significant increase in the number of water facilities.

However, the R^2 value of 0.0169 suggests that the reservation policy explains only about 1.7% of the total variance in the outcome. While statistically significant, the effect size is small relative to the overall variation in the data.